LOGICAL REASONING

Directions (Q. Nos. 1 & 2) Are based on the following A boy is asked to put in a basket one mango when ordered 'One', one orange when ordered "Two'. One apple when ordered "Three' and is asked to take out from the basket one mango and an orange when ordered 'Four'. A sequence of order is given as 12332142314223314113234

1. How many total fruits will be in the basket at the end of the above order sequence?
   (a) 9 (b) 8 (c) 11 (d) 10
2. How many total oranges were in the basket at the end of the above the sequence?
   (a) 1 (b) 4 (c) 3 (d) 2

Directions (Q. Nos. 3 to 6) : Are based on the following: Eight friends J, K, L, M, N, O, P and Q live on eight different floors of a building but not necessarily in same order. The lower most floor of the building is numbered one, the above that numbered two and soon until the lop most floor is numbered eight.

- J lives on floor numbered six.
- Only one person lives between J and L.
- O lives on the floor immediately below L.
- Only one person lives between O and P.
- O lives above P.
- K lives on an even numbered floor but not on the floor numbered two.
- Two persons live between K and Q.
- Q does not live on the lower most floor
- N lives on one of the floor above Q.

3. If P and L Interchange their places, who will live between P and M?
   (a)O (b)L (c)J (d) No one
4. Which of the following is True about 'M'
   (a) K lives immediately above M
   (b) Only two people live between M and Q
   (c) M lives on an odd numbered floor
   (d) M lives on the lower most floor
5. Who amongst the following lives on the floor number eight?
   (a) P (b) O
   (c) K (d) Cannot be determined
6. Three of the following four are alike in a certain way based on the given arrangement and thus form a group. Which of the following does not belong to the group?
   (a) PL (b) MQ
   (c) LN (d) OM

Directions (Q. Nos. 7 to 9): Are based on the followng A team must be selected from the ten probable players A, B, C, D, E, F, G, H, I and J. Of these, A, C, E and J are forwards, B, G and H are point guards and D, F and I are defender.

- The team must have at least one forward, one point guard and one defender.
- If the team includes J, it must also include F.
- The team must include E or B, but not both.
- If the team includes G, it must also include F.
- The team must include exactly one among C, G and I.
- C and F cannot be members of the same team.
- D and H cannot be members of the same team.
- The team must include both A and D or neither of them.
- There is no restriction on the number of members in the team.

7. What would be the size of the largest possible team?
   (a)7 (b)6 (c)5 (d)4
8. Which of the following cannot be included in a team of size 6?
   (a)A (b)H (c)J (d)E
9. What could be the maximum size of the team that includes G?
   (a)4 (b)5
   (c)6 (d)More than 6
10. A family has several children. Each boy in this family has as many sisters as brothers, but each girls has twice as many brothers as sister. How many brothers and sister are there?
    (a)1 and 2 (b)3 and 4
    (c)6 and 3 (d)4 and 3
11. In a certain code language'134' means 'Good and tasty', '478' means ' see good picture' and '729' means 'picture are faint'. Which of the following numerical symbols stand for see?
    (a)2 (b)7 (c)8 (d)1
12. If the English word 'EXAMINATION' is coded as 56149512965, then word 'GOVERNMENT' can be coded as
    (a)7655955552 (b)7645954452
    (c)7645954552 (d)7644956552
13. **Fact 1 :** Most stuffed toys are stuffed with beans.
    Fact 2 : There are stuffed bears and stuffed tigers.
    Fact 3 : Some chair are stuffed with beans.
    If the above statement are fact. Which of the following statements, must also be fact?
    1. Only children's chairs are stuffed with beans.
    2. All stuffed tigers are stuffed with beans.
    3. Stuffed monkeys are not stuffed with beans.
    (a) 1 is fact
    (b) Only 2 is fact
    (c) Both 2 and 3 are facts
    (d) None of the statements 1, 2, 3 are true
14. If all the 6' s are replaced by 9's then the algebraic sum of all the numbers from 1 to 100 (both inclusive) varies by
    (a) 333 (b) 300 (c) 279 (d) 330
15. 405 sweets were distributed equally among a group of children such that the number of sweet received by each child is one fifth of the number of children. The number of children in the group is

(a) 45 (b)9 (c)21 (d)15

16. The number of common terms in the two sequence 17, 21,25, …….. ,817 and 16, 21,26,…. ,851 is
(a)28 (b)39 (c)40 (d)87

17. Unscramble the letter of the following words and find the odd one.
(a)ONGEAR (b)NOONI
(c)ALPEP (d)AUVAG

18. A bus starts from its depot filled to seating capacity. It stops at a point A where 1/6th the of passengers alight and 10 board the bus. At point B, 1/5th of the passengers alight and boards the bus. At point C which is the last stop, all the 55 passengers alight. The capacity of the bus is
(a) 96 (b) 99 (c) 66 (d) 90

19. Kha- kha is an obscure island which is inhabited by two types of people; the 'Yes' type and the 'No' type. Native of types 'Yes' ask only questions the right answer in which is 'Yes' while those of type 'No' ask only questions the right answer to which is 'No'. For example the 'Yes' type will ask questions like "Is 2 plus 2 equal to 4?" While the "No" type will ask question like "Is 2 plus 2 equal to 5?". The following question is based on your visit to the island Kha-kha. Kevin and Kumar are brothers from the island. Kumar asks you, Is atleast one of us is of type 'No'? You can conclude that
(a)Kevin is 'No', Kumar is 'Yes'.
(b)Both are 'Yes'.
(c)Kevin is 'Yes', Kumar is 'No'.
(d)Both are 'No'.

20. Raman was born on March 5, 1970. Lakshman was born 25 days before Raman . The year when they took birth. the Republic Day fell on Monday. What is the day of birth of Lakshman.
(a) Sunday (b) Monday
(c) Wednesday (d) Saturday

21. P, Q, R and S are four logical statements such that if P is true, then Q is true; if Q is true, then R is true; and if S is true, then at least one of Q and R is false. The it follows that
(a) if S is false, then both Q and R true
(b) if at least one of Q and R is true, then S is false
(c) if P is true, then S is false
(d)if Q is true, then S is false

22. A, B, C, D, E, F and G are members of a members of a family consisting of four adults, three children, three males and four female. Out of the children F and G are girls. A and D are brothers and A is doctor, E is an engineer married to one of the brothers and has two children . B is married to D and G is their child. Who is C?
(a) E's daughter (b) A's son
(c) G's brother (d) F's father

23. A clock is set right at 5 am the clock losses 16 minutes in 24 hours. What will be the correct time when the clock indicates 10 pm on the 3rd day?
(a)11 pm (b)10: 45 pm
(c)11: 15 pm (d)12 pm

24. The day after the day after tomorrow is four days after Monday. What day is it today?
(a) Monday (b)Tuesday
(c) Wednesday (d)Thursday

Directions (Q. Nos. 25 to 27) :

- There are six houses P, Q, R, S, T and U, three on either side of a road.
- The houses are of different colours - red, blue, green, orange, yellow and white.
- All the houses are of different heights.
- T, the tallest house is exactly opposite to the red coloured house.
- The shortest house is exactly opposite to the green coloured house.
- U, the orange coloured house is located between P and S.
- R, the yellow coloured house is exactly opposite to P.
- Q, the green coloured house is exactly opposite to U.
- P, the white coloured house is taller than R, but shorter than S and Q.

25. Which is the second largest house?
(a) Q (b) R
(c) S (d) Cannot be determined

26. Which of the second shortest houses?
(a) P (b)R
(c)S (d)Cannot be determined

27. What is the colour of the tallest house?
(a)Red (b)Blue
(c)Green (d)Yellow

28. How many pairs of letters are there in the word NECESSARY which have as many letters between them in the word as there are between them in the alphabet and in the same order.
(a) One (b) two
(c) three (d) four

29. What is missing number is the series, 4, 7, 11, 18, 29, 47,…….. , 123, 199 ?
(a)76 (b)77 (c)86 (d)87

Directions (Q. Nos. 30 to 32) :

- A, B, C, D, E and F are six members in a family in which there are two married couples.
- D is brother of F.
- Both D and F are lighter than B.
- B is mother of D and lighter than E.
- C a lady, is neither heaviest nor lightest in the family.
- E is lighter than C.
- The grandfather in the family is the heaviest.

30. Which of the following is a pair of married couples.
(a)AB (b)BC
(c)AD (d)BE

31. Who among the following will be in the second place, if all the members in the family are arranged in a descending order of their weights?

(a)C (b)A
(c)D (d)Data inadequate

32. How is C related to D?
(a)Grandmother (b)Cousin
(c)Sister (d)Mother

Directions (Q. Nos. 33 to 35) : are based on the following:

- Eleven students, A, B, C, D, E, F, G, H, I, J and K are sitting in the first row of the class facing the teacher.
- D who is to the immediate left of F is second to the right of C.
- A is second to the right of E, who is at one of the ends.
- J is the immediate neighbor of A and B and third to the left of G.
- H is to the immediate left of D and third to the right of I.

33. Who is sitting in the middle of the row?
(a)B (b)C (c)G (d)I

34. If E and D, C and B, A and H and K and F interchange their position, which of the following pairs of students are sitting at the ends?
(a)D and E (b)E and F
(c)D and K (d)K and F

35. Which of the following groups of friends is sitting to the right of G?
(a)CHDE (b)CHDF
(c)IBJA (d)None of the above

Directions (Q. Nos. 36 to 39)are based on the following:

- A group of six friends are sitting around a hexagonal .table, each one at one corner of the hexagon.
- Ram is sitting opposite to Ramesh.
- Jyoti is sitting next to Seema.
- Neeta is sitting opposite to Seema but not next to Ram.
- Amrit has a person sitting between Ramesh and himself.

36. If Seema and Jyoti mutually interchange their positions, then who will be sitting opposite to Neeta?
(a)Jyoti (b)Ram
(c)Seema (d)Ramesh

37. If Neeta sits to the right of Amrit, then who is sitting to the left of Amrit?
(a)Ramesh (b)Neeta
(c)Jyoti (d)Ram

38. Who is sitting between Amrit and Ramesh?
(a)Neeta (b)Jyoti
(c)Seema (d)Ram

39. Who is sitting opposite to Jyoti?
(a)Ramesh (b)Neeta
(c)Amrit (d)Seema

40. A Group of 630 children are seated in n rows for a group photo session. Each row contains three less children than the row in front of it. Which one of the following number of rows is not possible?
(a)3 (b)4 (c)5 (d)6

ANSWER

1.	C	2.	D	3.	A	4.	B	5.	C
6.	D	7.	A	8.	B	9.	C	10.	D
11.	C	12.	C	13.	D	14.	D	15.	A
16.	C	17.	B	18.	C	19.	A	20.	A
21.	C	22.	B	23.	B	24.	B	25.	D
26.	B	27.	B	28.	A	29.	A	30.	D
31.	A	32.	A	33.	D	34.	C	35.	B
36.	A	37.	D	38.	A	39.	C	40.	d

SOLUTION

1. (c) Total number of fruits in the basket
$= 19 \times 1 - 4 \times 2$
$= 19 - 8 = 11$

2. (d) Total number of two's = total number of orange = 6
Total number of four = 4
∴ Total number of oranges in the basket
$= 6 - 4 = 2$

Solutions (Q. Nos. 3- 6)

From the given information the order will be as

8	K
7	N
6	J
5	Q
4	L
3	O
2	M
1	P

3. (a) After interchanging the places, O lives between P and M.

4. (b) Only two people live between M and Q.

5. (c) K lives on floor number 8.

6. (d) Only O and M are adjacent to each other.

Solutions (Q. Nos. 7- 9)

Forwards - A, C, E and J
Point Guard - B, G and H
Defenders - D, F and I

7. (a) The largest possible team shall be of 6.
F, J, A, D, G and B/E.

8. (b) H cannot be included in 6 member team as if H is included then A and D will not be in the team.

9. (c) G will be a member of 6 member team.

10. (d) Through option we can check the possibilities. With option (d), we have
B – B – B – B
S – S – S
So, each boy has 3 sisters and 3 brothers and each girl has 2 sisters and 4 brothers.

11. (c)

13	4		-	good and tasty
4	7	8	-	see good picture
7	29		-	picture are faint

As, good = 4
Picture = 7
So, see = 8

12. (c) As, E → 05 → 0 + 5 = 5
X → 24 → 2 + 4 = 6
A → 01 → 0 + 1 = 1

Similarly,

M	→	13	→	1	+	3	=	4
I	→	09	→	0	+	9	=	9
N	→	14	→	1	+	4	=	5
A	→	01	→	0	+	1	=	1
T	→	20	→	2	+	0	=	2
I	→	09	→	0	+	9	=	9
O	→	15	→	1	+	5	=	6
N	→	14	→	1	+	4	=	5
G	→	= 07	→	0	+	7	=	7
O	→	15	→	1	+	5	=	6
V	→	22	→	2	+	2	=	4
E	→	05	→	0	+	5	=	5
R	→	18	→	1	+	8	=	9
N	→	14	→	1	+	4	=	5
M	→	13	→	1	+	3	=	4
E	→	05	→	0	+	5	=	5
N	→	14	→	1	+	4	=	5
T	→	20	→	2	+	0	=	2

GOVERNMENT-7645954552

13. (d) None of the given statements match the facts.

14. (d) There are 19 numbers which has 6 as a digit when unit digit 6 is replaced by 9, then sum of numbers will be increase by = 3 × 10 = 30
From 60 to 69 there are 10 numbers which have ten's digit 6.
Here, the increase will be 30 × 10 = 300
∴ Total increase = 30 + 300 × 330

15. (a) Let the number of children be x.
∴ $x \times \frac{x}{5} = 405$
$\Rightarrow x^2 = 5 \times 405 = 2025$
$\Rightarrow x = \sqrt{2025} = 45$

16. (c) Number of terms in
1st series 817 = 17 + (m – 1)4
⇒ m = 201
2nd series, 851 = 16 + (n – 1)5
⇒ n = 168
Now, 17 + (m – 1)4 = 16 + (n – 1)5
⇒ 4m + 2 = 5n
For, m = 2, n = 2
m = 7, n = 6
m = 12, n = 10
m = 17, n = 14
It also form a series.
∴ Number of common terms.
197 = 2 + (P – 1) 5
P = 40

17. (b) (a) ONGEAR - ORANGE
(b) NOONI - ONION
(c) ALPEP - APPLE
(d) AUVAG - GUAVA
Onion is the only thing that is taken out of the soil.

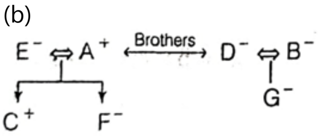
18. (c) At C, number of passenger = 55
∴ Total number of passenger at B = $\frac{5}{4} \times (55 - 3)$
$= \frac{5}{4} \times 20 = 65$
∴ Total number of passengers at
A = $\frac{6}{5} \times (65\text{-}10)$
$= \frac{6}{5} \times 55 = 66$

19. (a) If the answer of question is 'No', then none belongs No-zone. This implies both are Yes. Which is contradiction.
If Kumar is 'Yes' the answer to the questions is 'Yes' tha implies that Kevin is from No.

20. (a) Raman's DOB = 5th March, 1970
∴ Lakshman's DOB = 8th Feb 1970
Republic Day, i.e. 26 Jan = Monday
∴ 8th Feb = Sunday.

21. (c) If P is true, then Q is true.
If Q is true, then R is true.
So, if P is true, then Q and R are both true.
and if Q and R are true, then S is false.

22. (b)

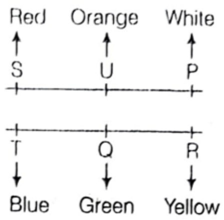


So, C is son of A and E.

23. (b) Total hours from 5 am to 10 pm of 3rd day = 65h
The clock looses 16 min in 24 h.
i.e. 23 h 44 min of the clock = 24h of the correct clock
So, 65 h of clock = $\frac{24 \times 15 \times 65}{356}$ of the correct clock
= 65.73h
= 65h 44 min
So, correct time = 10:45pm (approx)

24. (b) 4 days after Monday is Friday
So, today is Tuesday.

Solutions. (Q. Nos. 25-27)
From the given information the arrangement will be as follows.



Order of height T > S/Q > P >R>U
25. (d) 2nd highest cannot be determined.
26. (b) 2nd shortest house is R.
27. (b) The tallest house is T which is Blue.

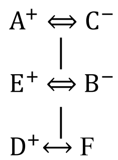
28. (a)
NECESSARY
There is only one such pair NS.

29. (a) The series is as follows

4				
7				
7	+	4	=	11
11	+	7	=	18
11	+	18	=	29
18	+	29	=	47
29	+	47	=	$\boxed{76}$
47	+	76	=	123
76	+	123	=	199

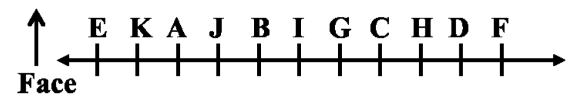
∴ Missing term = 76

Solutions (Q. Nos. 30-32) From the given information,



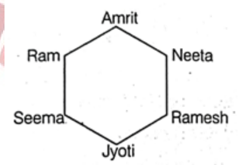
According to weight A>C>E>B>D,F
30. (d) BE is a couple
31. (a) C will be at 2nd place.
32. (a) C is grandmother of D.

Solutions (Q. Nos. 33-35) From the given information the arrangement will be as



33. (d) I is sitting in the middle of the row.
34. (c) D and K will be at end points.
35. (b) CHDF are sitting to the right of G.

Sol. (Q.Nos. 36-39) From the given information the arrangement is as follows.



36. (a) After interchanging their position Jyoti will be opposite to Neeta.
37. (d) Ram is sitting left to Amrit.
38. (a) Neeta is sitting between Amrit and Ramesh.
39. (c) Amrit is sitting opposite to Jyoti.
40. (d) We can check it through option.